

ROAD SAFETY HACKATHON 2026 • IIT MADRAS

RoadSoS

The AI Companion that turns every bystander into a first responder

Saving lives in the Golden Hour through AI-driven triage, location-aware dispatch, and voice-first accessibility in every Indian language.



Leon Kaushik Deka • Team RBG Labs • CoERS, IIT Madras • BIMSTEC Track

English

हिन्दी

অসমীয়া

বাংলা

தமிழ்

ಕನ್ನಡ

+ more

THE PROBLEM

1.19M deaths/year.
Half within 60 minutes.

METRIC	GLOBAL	BIMSTEC
Ambulance response (urban)	8–12 min	20–35 min
Ambulance response (rural)	15–25 min	45–90+ min
Bystander first-aid rate	15–25%	< 5%
Deaths within Golden Hour	~50%	60–70%

"Most aren't killed by the crash. They're killed by the wait."

OUR SOLUTION

A multi-modal AI emergency companion — **WhatsApp, voice, SMS, AR** — that orchestrates the entire chain of survival.



SPEED

One tap or one phrase. No login walls during an emergency. Crash-detection ready.



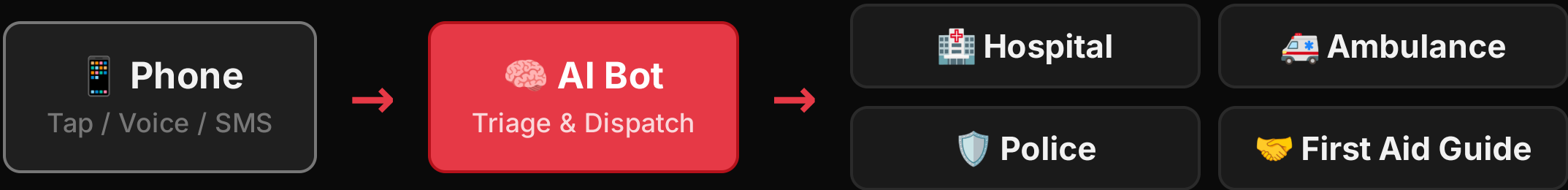
VERNACULAR

English, Hindi, Assamese, Bengali, Tamil and more — voice-first, with code-mixed speech understood natively.



RESILIENCE

Offline-first design with SMS fallback. One bar of GSM signal is enough to dispatch help.



HOW IT WORKS

11 PM on NH-37, Assam — a real incident, second by second.

0:00 TRIGGER Ravi taps the SOS button — or simply sends "SOS" on WhatsApp.	0:05 LOCATE GPS locked to 16 m. First triage question asked — in the user's language: Assamese, Hindi or English.
0:20 DISPATCH Classified HIGH severity → an ALS ambulance is dispatched — not the closer PHC that lacks trauma care.	0:35 FIRST AID Voice guide walks Ravi through bleeding control, with a live AR camera overlay.
1:10 NOTIFY Nearest police station alerted; the victim's ICE emergency contact is called automatically.	11:20 GOLDEN HOUR Ambulance arrives. The ER is pre-warned with triage notes. The chain of survival held.

BUILT WITH

LangGraph

Llama 3.1

FastAPI

Whisper ASR

IndicTrans2

PostGIS

WhatsApp API

pgvector

MediaPipe Pose

The triage question in panel 2 is generated and spoken in whatever language the caller uses — Assamese on NH-37, Hindi on NH-44, English in a metro. The same flow runs everywhere.

Engineered for life-critical reliability

PRESENTATION

PWA | WhatsApp | IVR | SMS | AR



AI ORCHESTRATION

LangGraph Agent | RAG (WHO Protocols) | IndicTrans2 | Whisper ASR



SERVICES

Hospitals | Ambulances | Police | Towing | Puncture Shops



DATA

PostgreSQL + PostGIS | pgvector | Redis | Kafka

SAFETY-FIRST AI

RAG, not free generation — every instruction is grounded in the WHO / ATLS trauma corpus. If retrieval confidence falls below 0.78, the case is escalated to a human dispatcher.

Open-weights LLM

On-prem deployable

Offline-first

8 languages — English, Hindi, Assamese & more

From hackathon to BIMSTEC-wide deployment

22-
34%

mortality reduction — modelled against WHO trauma curves if accident-to-care response time is cut by just 15 minutes.

1

JUN 2026

Stage 1 submission (now) — PWA, WhatsApp bot, triage agent, multilingual.

2

Q3 2026

NH-44 + NH-66 pilot with state EMS partnerships.

3

Q4 2026

6 Indian states + Sri Lanka + Bhutan expansion.

4

2027

Cross-BIMSTEC rollout with national 112 integration.

Open-source SDK • Privacy-by-design • CoERS-led governance

THANK YOU

Every minute matters.
Let's give every road user a fighting chance.



TRY THE PWA

frontend-ten-gamma-78.vercel.app



API DOCUMENTATION

rare-adventure-production-984a.up.railway.app/docs

GitHub: github.com/leonkaushikdeka/roadsos

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